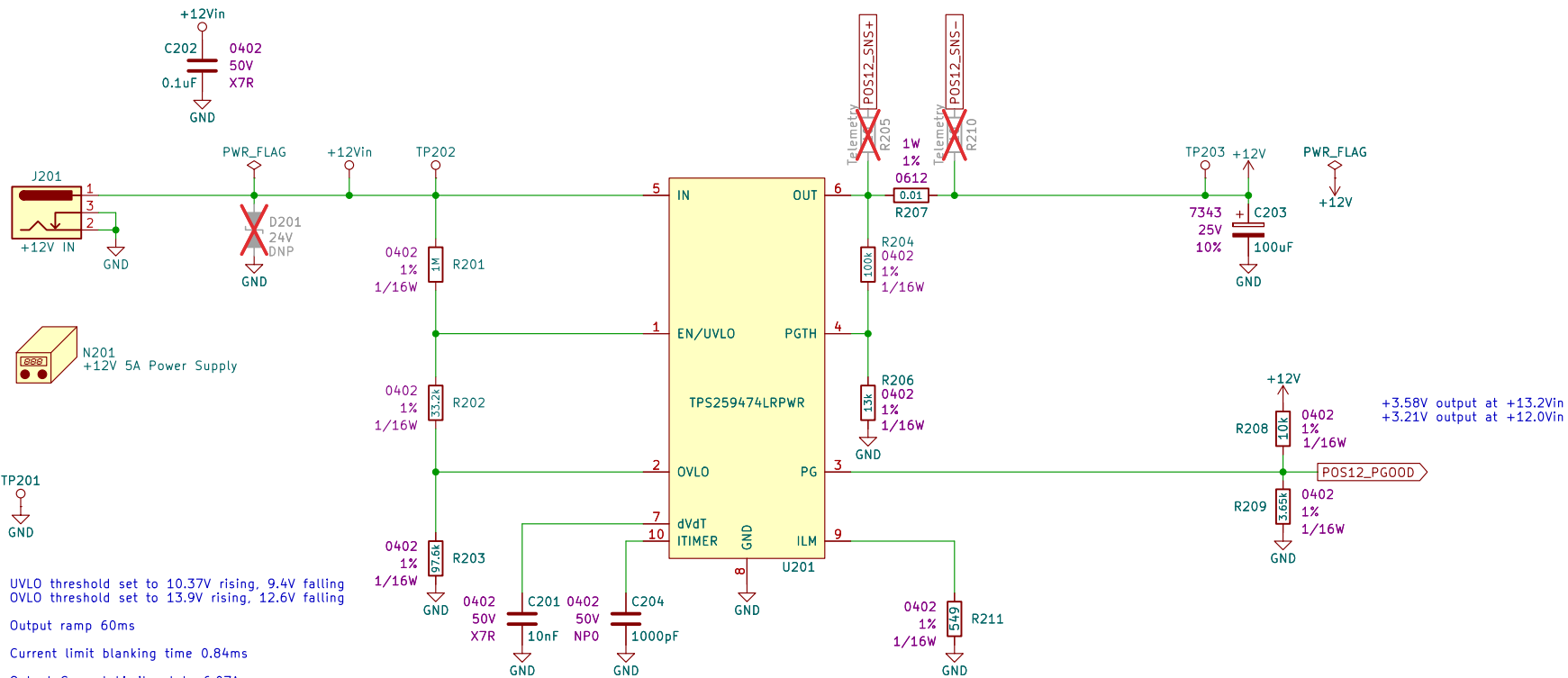


	1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch		Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Buffers File: led_panel_buffers.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch LED Panel Connectors File: led_panel_connectors.kicad_sch			
B						
C						
D						
	1	2	3	4	5	

Sheet: / File: Audio_Visualiser_DSP.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 1/21



UVLO threshold set to 10.37V rising, 9.4V falling
 OVLO threshold set to 13.9V rising, 12.6V falling
 Output ramp 60ms
 Current limit blanking time 0.84ms
 Output Current Limit set to 6.07A
 PGOOD threshold 10.43V rising, 9.5V falling

Load	Draw	Quantity	Total
LED Panel VLED 256 * 0.02A @ 3V = 15.36W / 12V = 1.28A / 80% efficiency	1.6A	2	3.2A
LED Panel POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	2	0.688A
Local POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	1	0.344A
Total		50.784W	4.232A

Sheet: /12V Power Input/
 File: POS12_power_input.kicad_sch

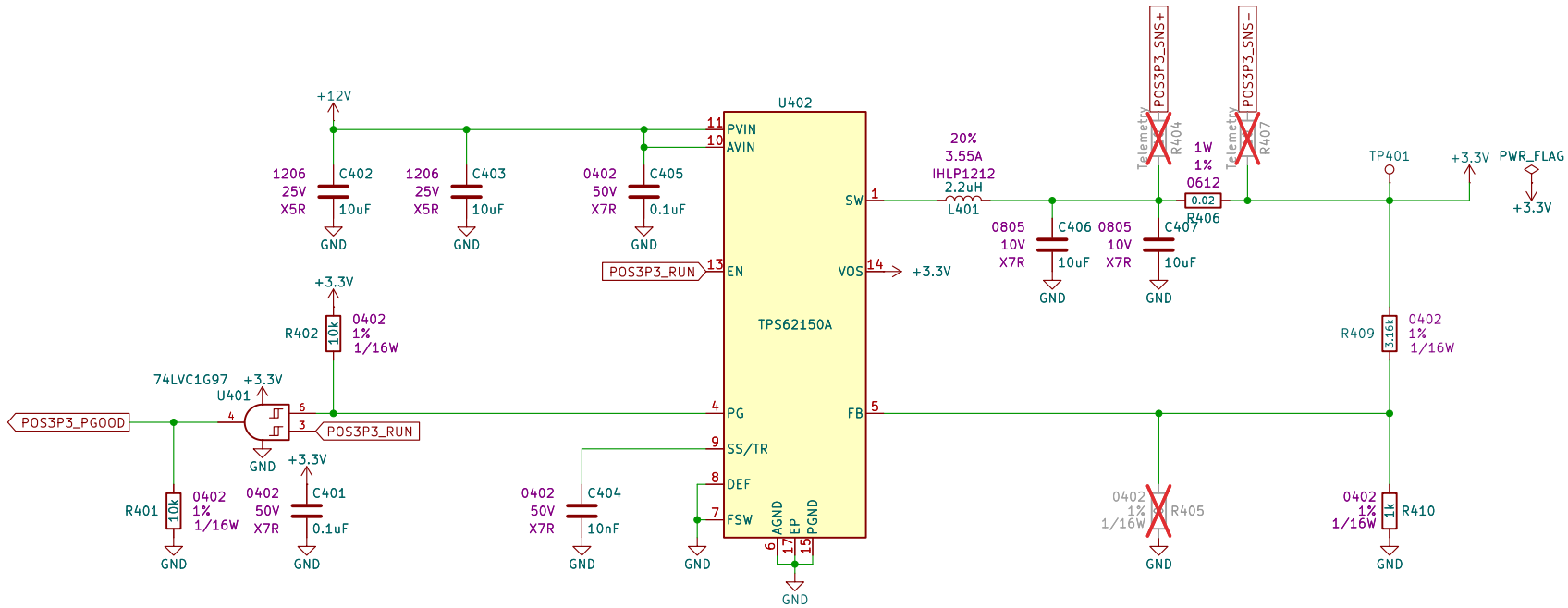
Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 2/21

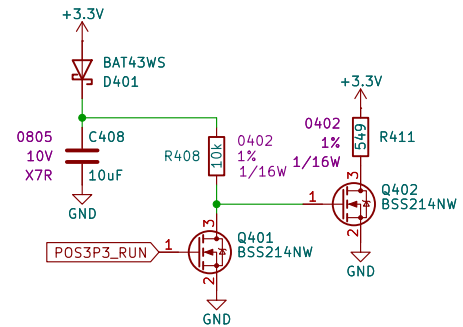


Rev:
Id: 3/21



Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	6	6.6mA
MCP6024	5.4mA	3	16.2mA
Analog_Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V_Reference	0.2mA	1	0.2mA
PCA9536_GPIO_Expander	0.29mA	1	0.29mA
I2C_Buffers	8.3mA	3	24.9mA
Octal_Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary_Encoder_Module	50mA	1	50mA
LEDs	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap_Touch_Sensor	0.729mA	1	0.729mA
Total		0.579W	175.48mA

POS12_PG00D 0 POS3P3_RUN
R403

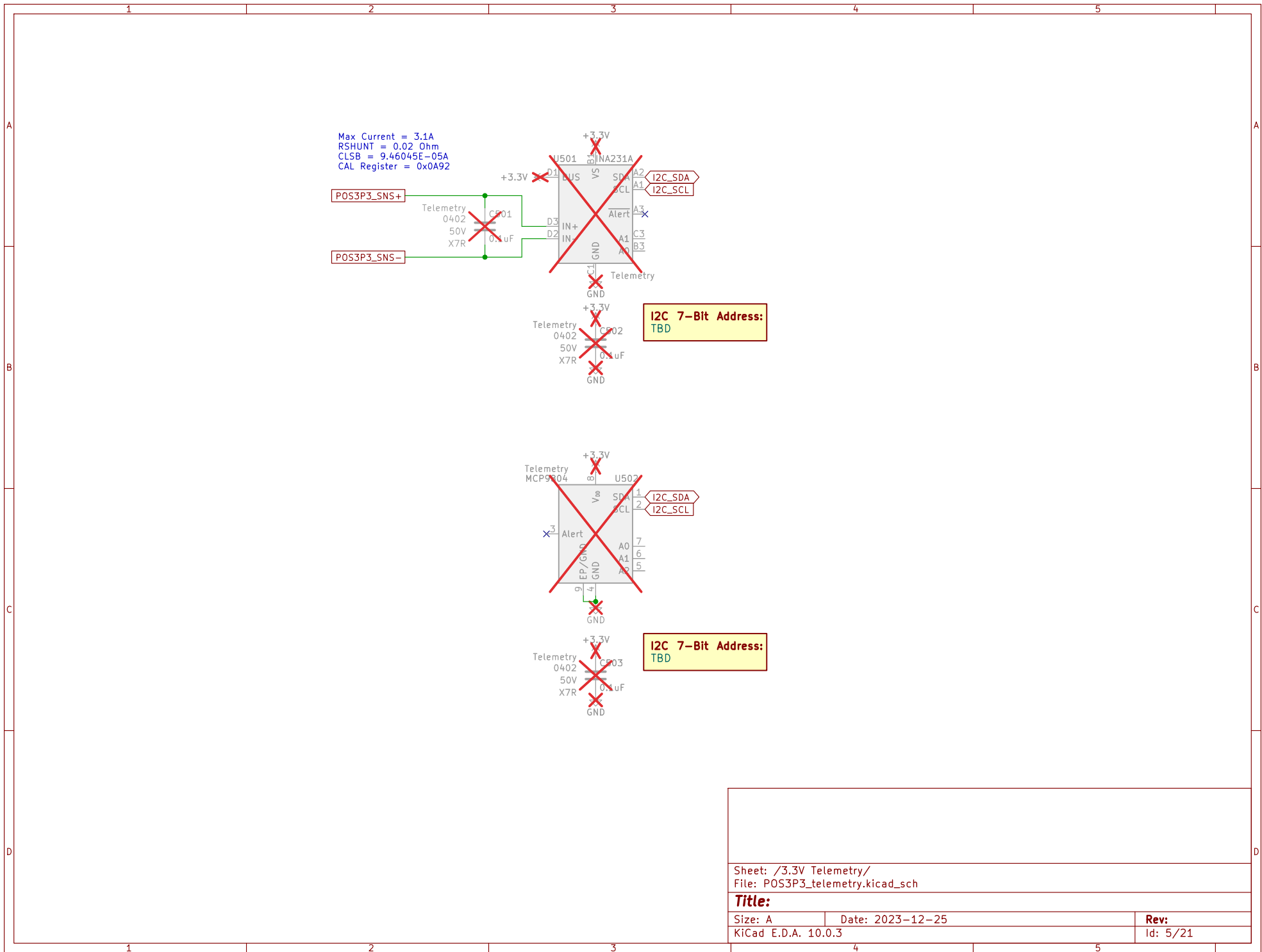


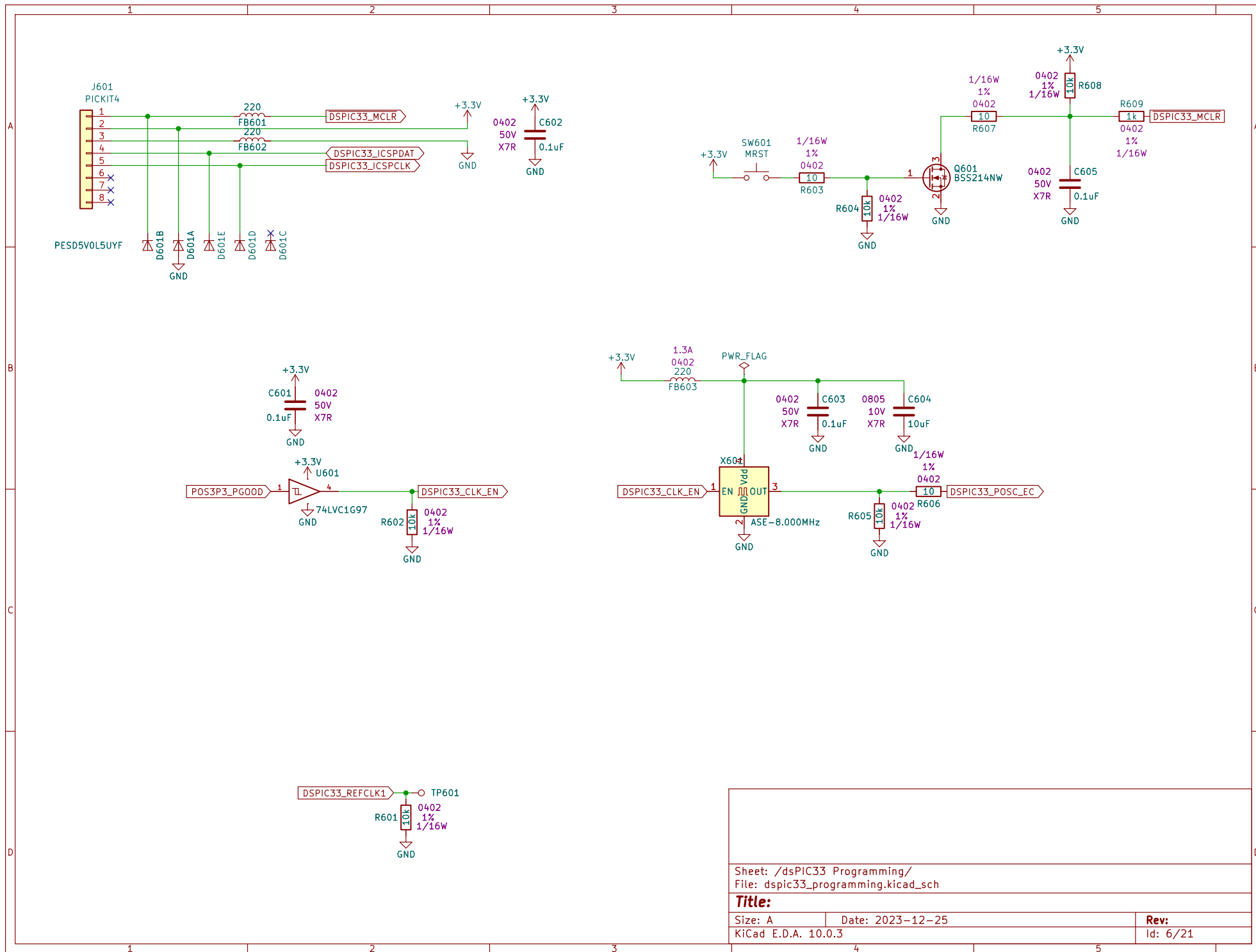
Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

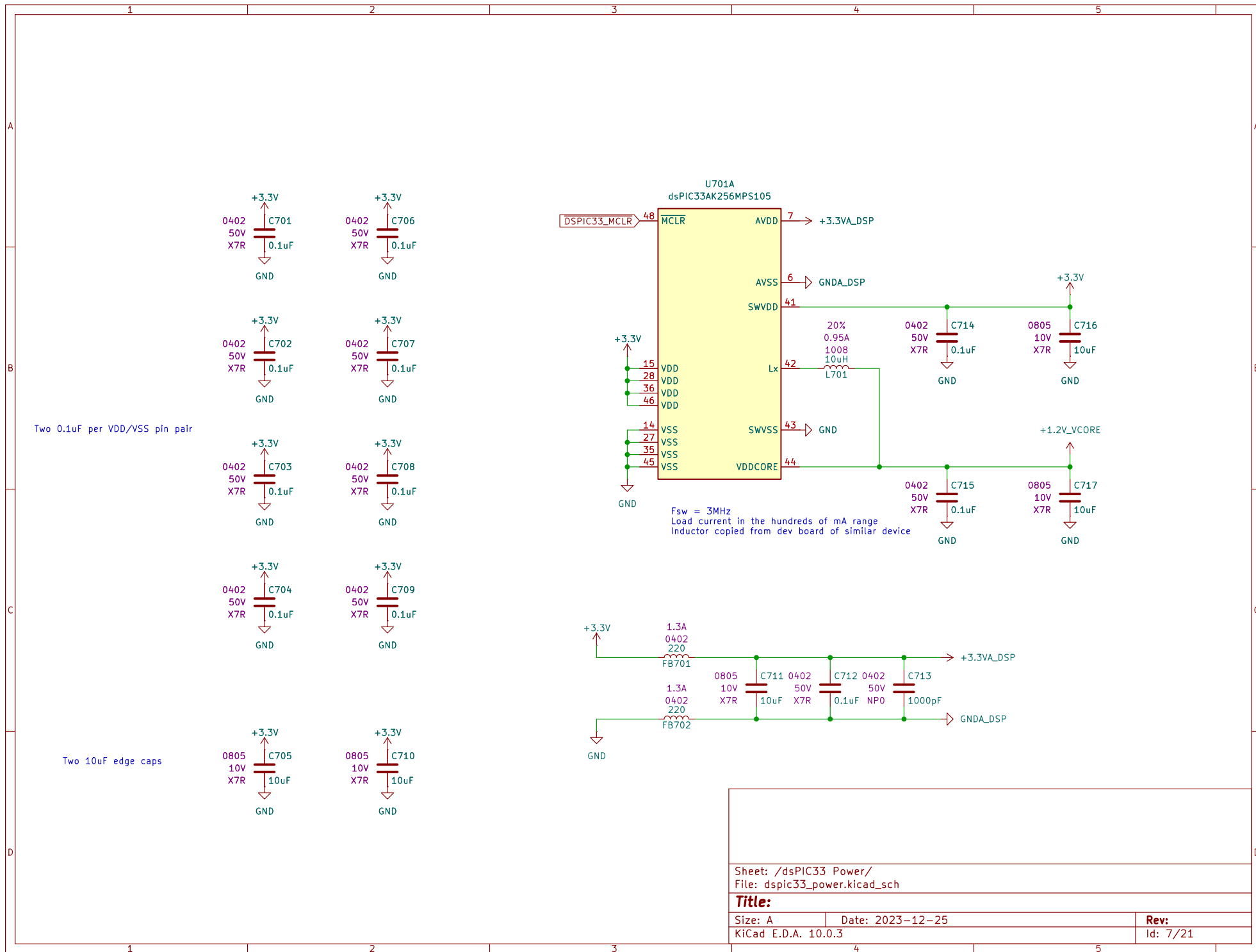
Title:

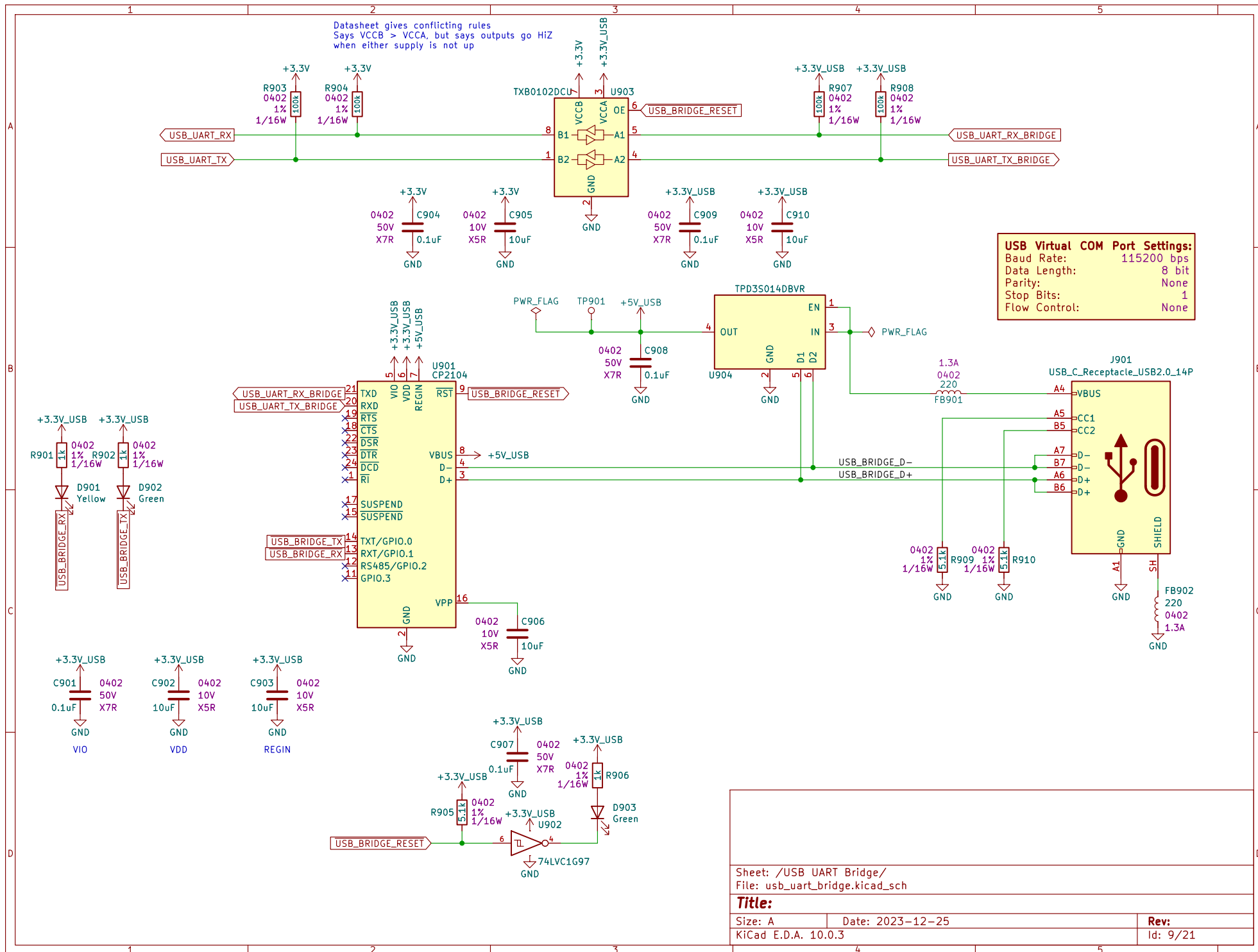
Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

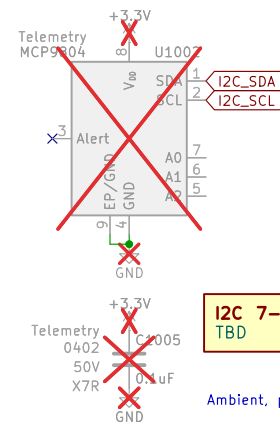
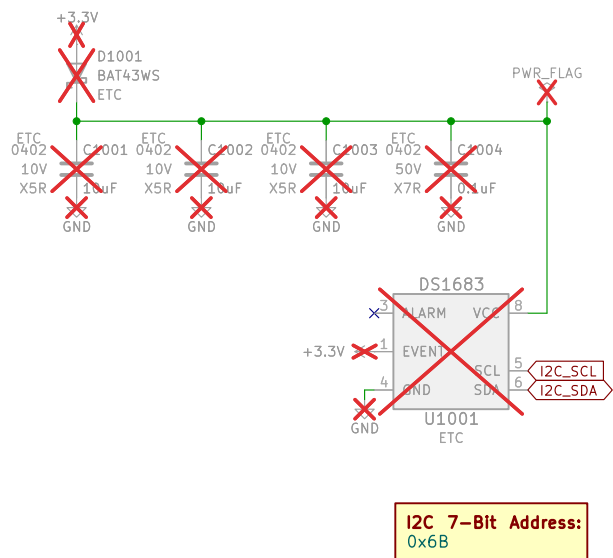
Rev:
Id: 4/21





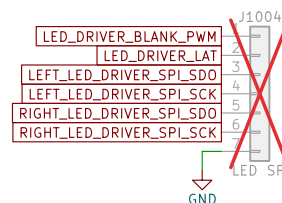
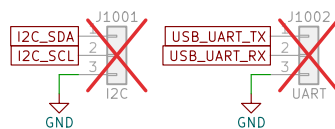






Ambient, place away from power components

Test Points

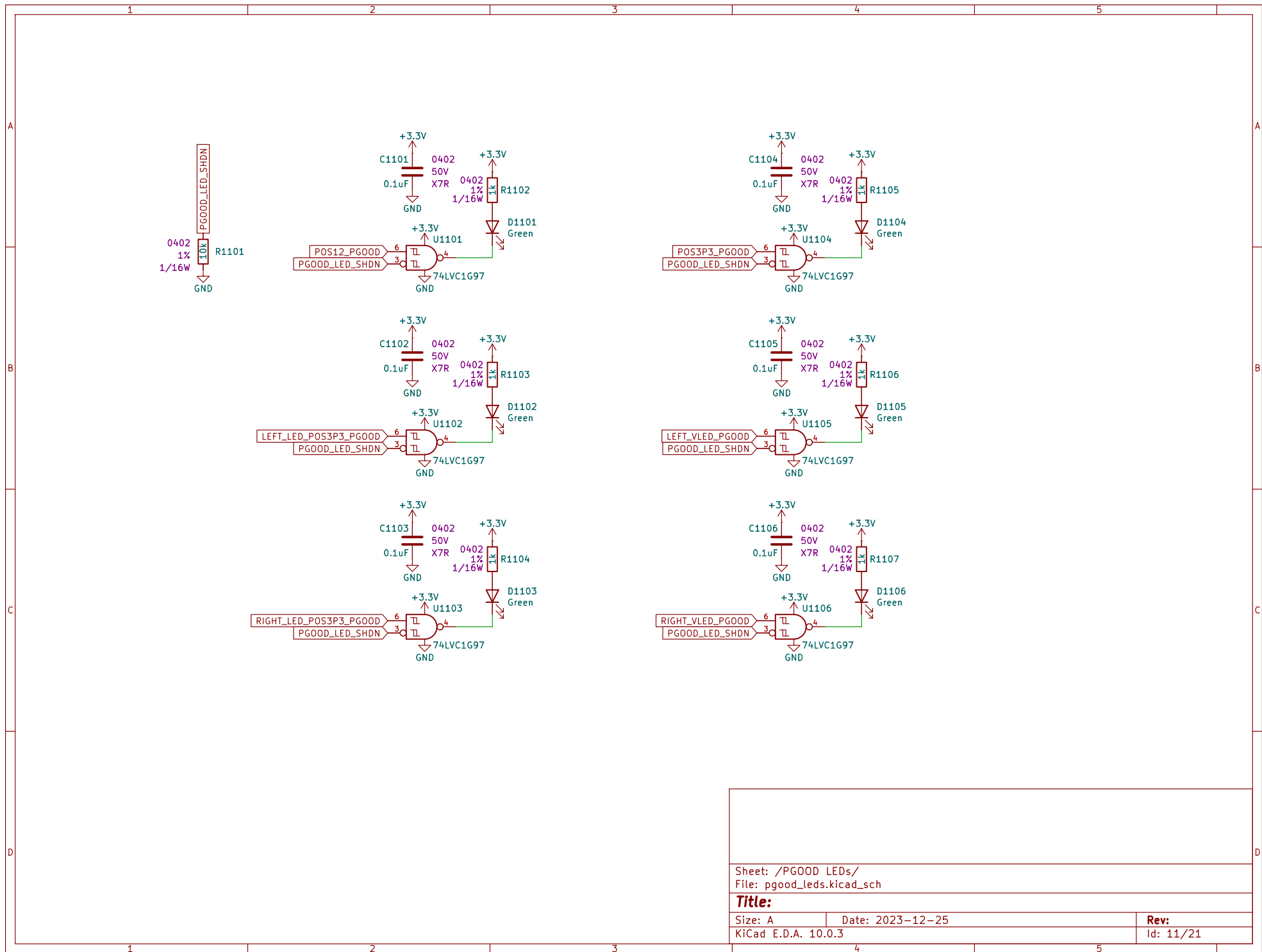


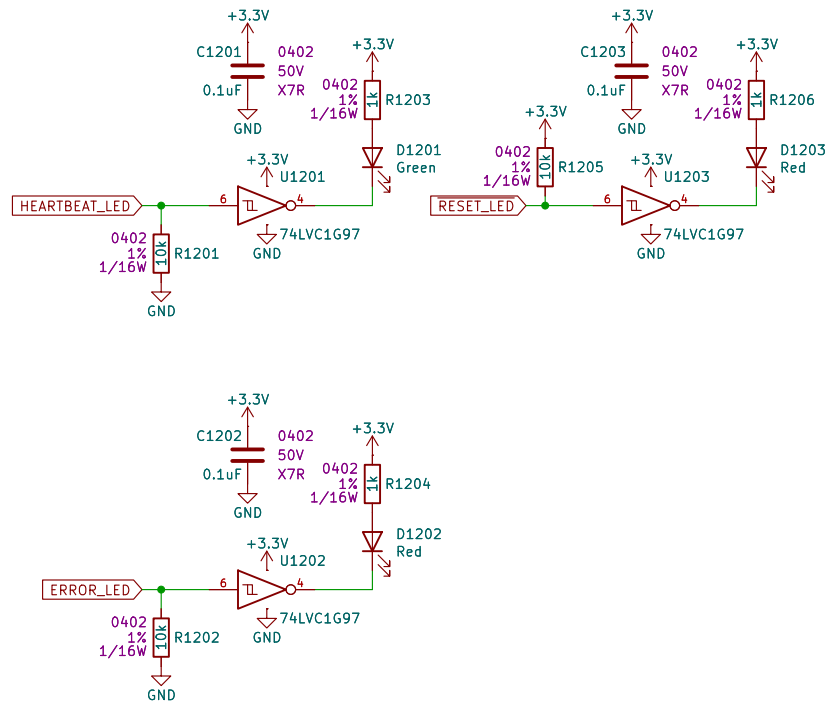
Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 10/21



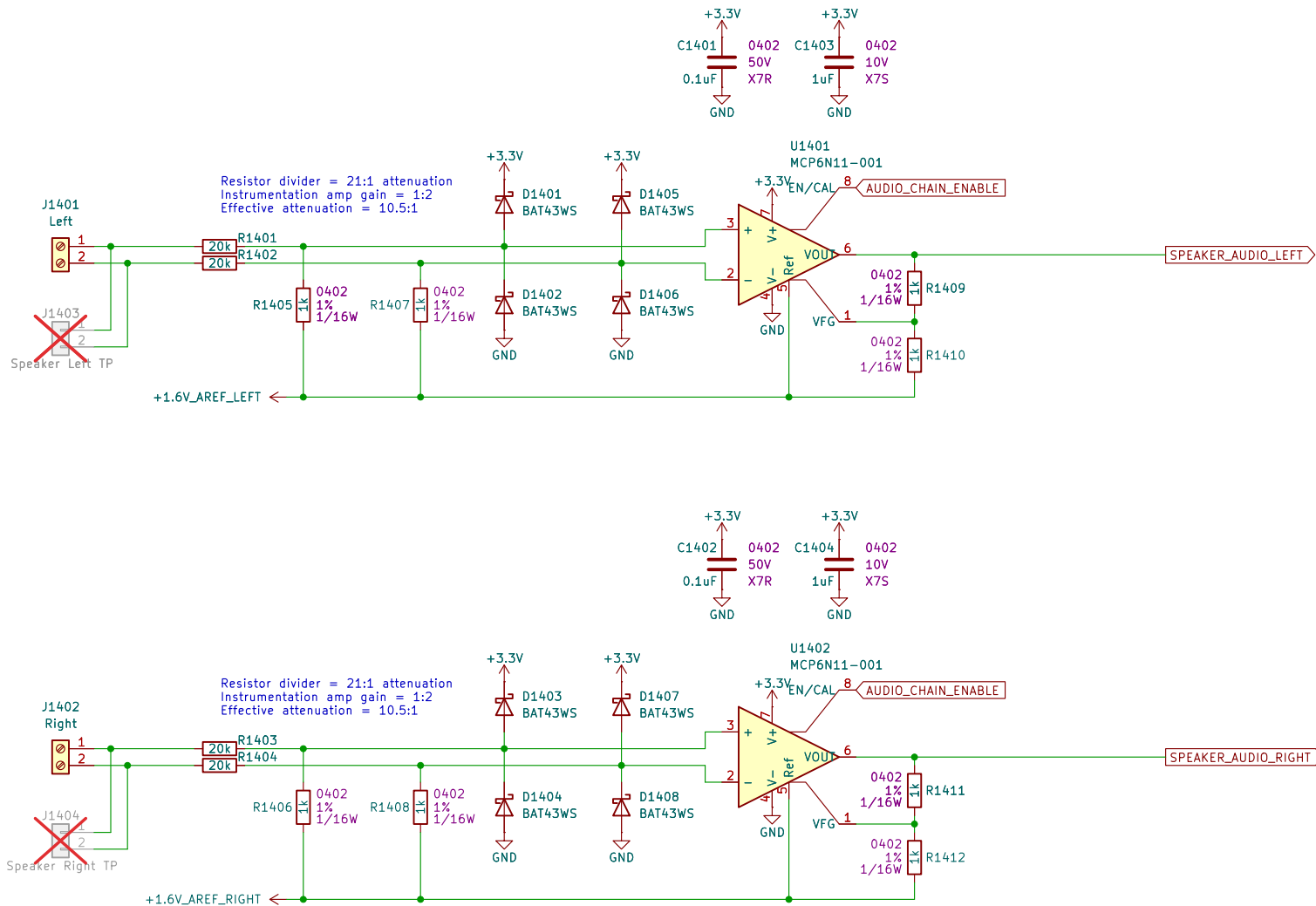


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/21



Sheet: /Speaker Signal Conditioning/
File: speaker_signal_conditioning.kicad_sch

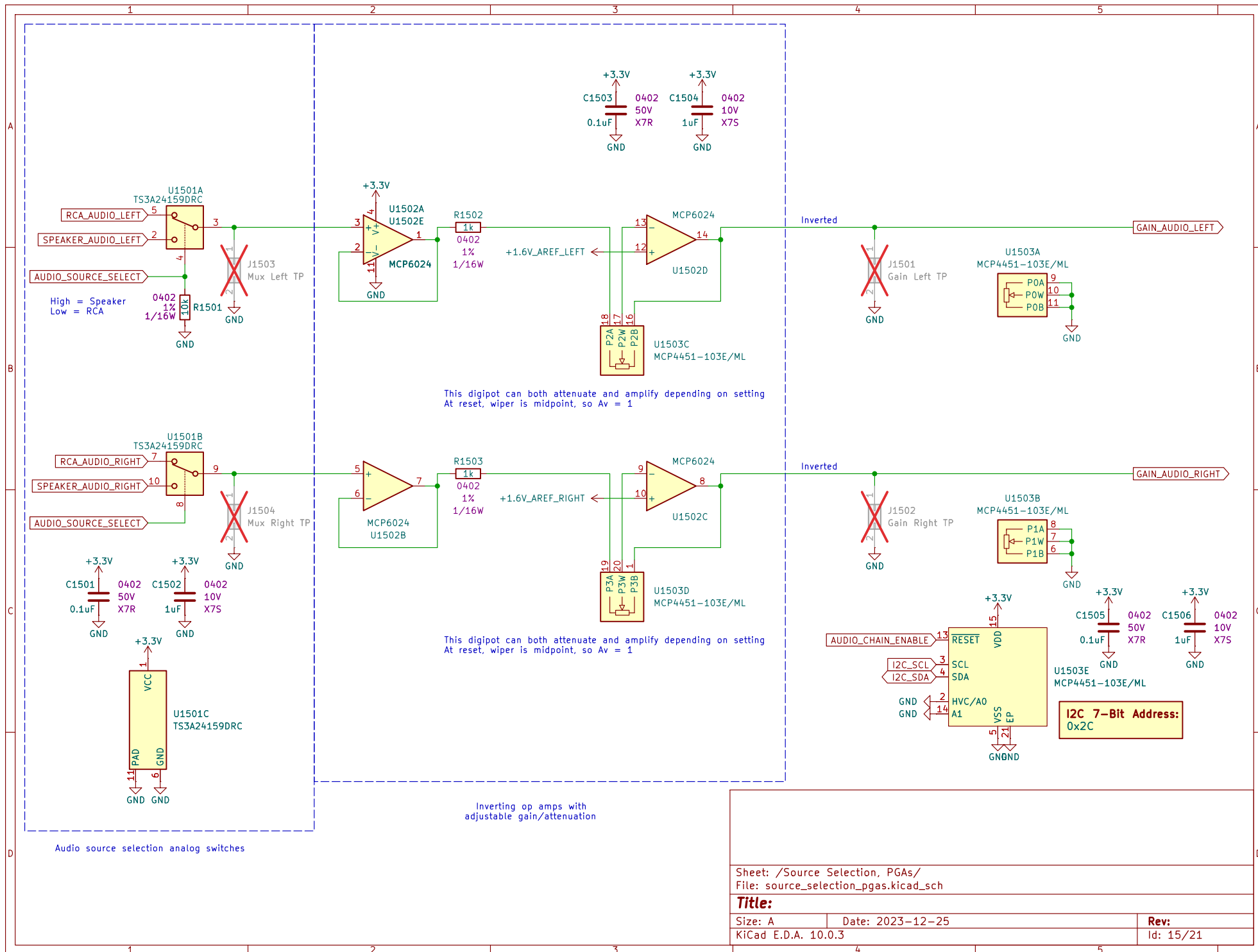
Title:

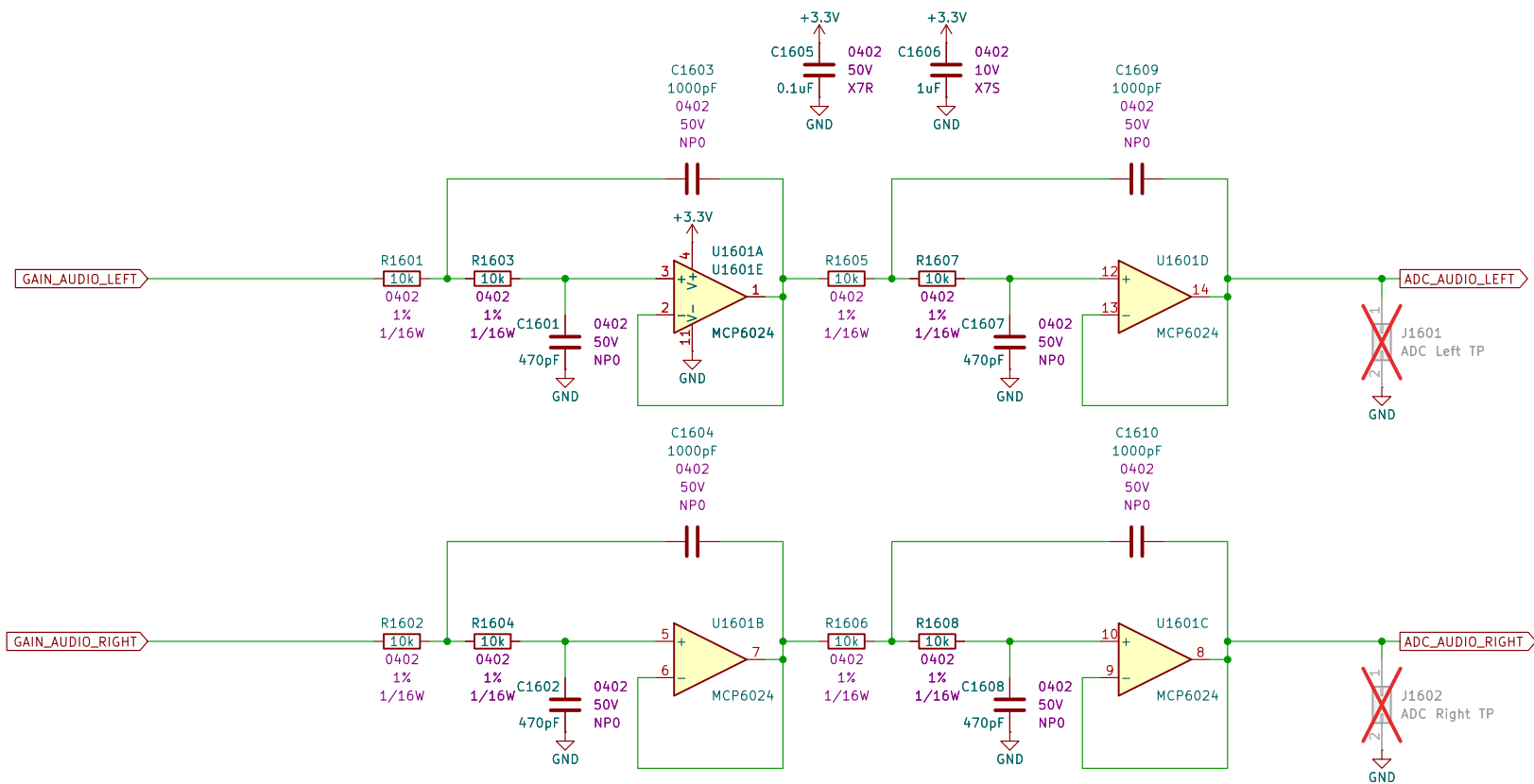
Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:

Id: 14/21



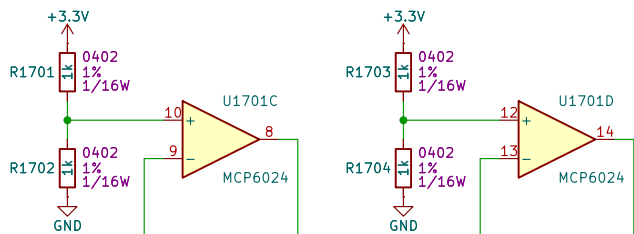
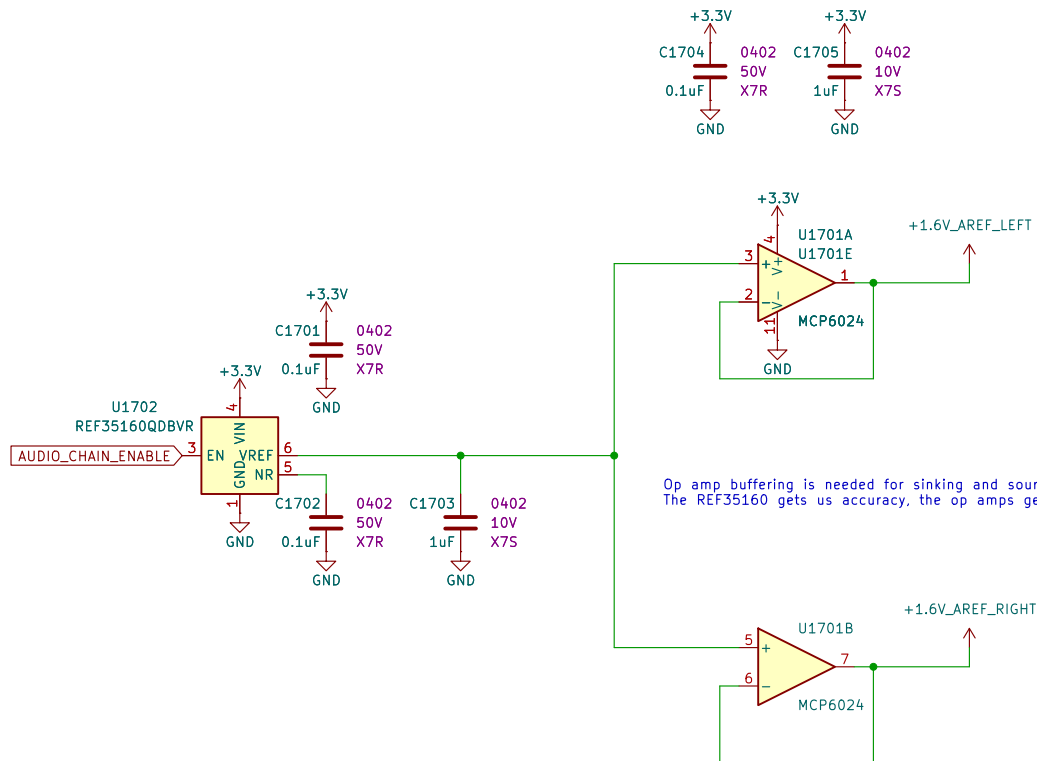


Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 16/21

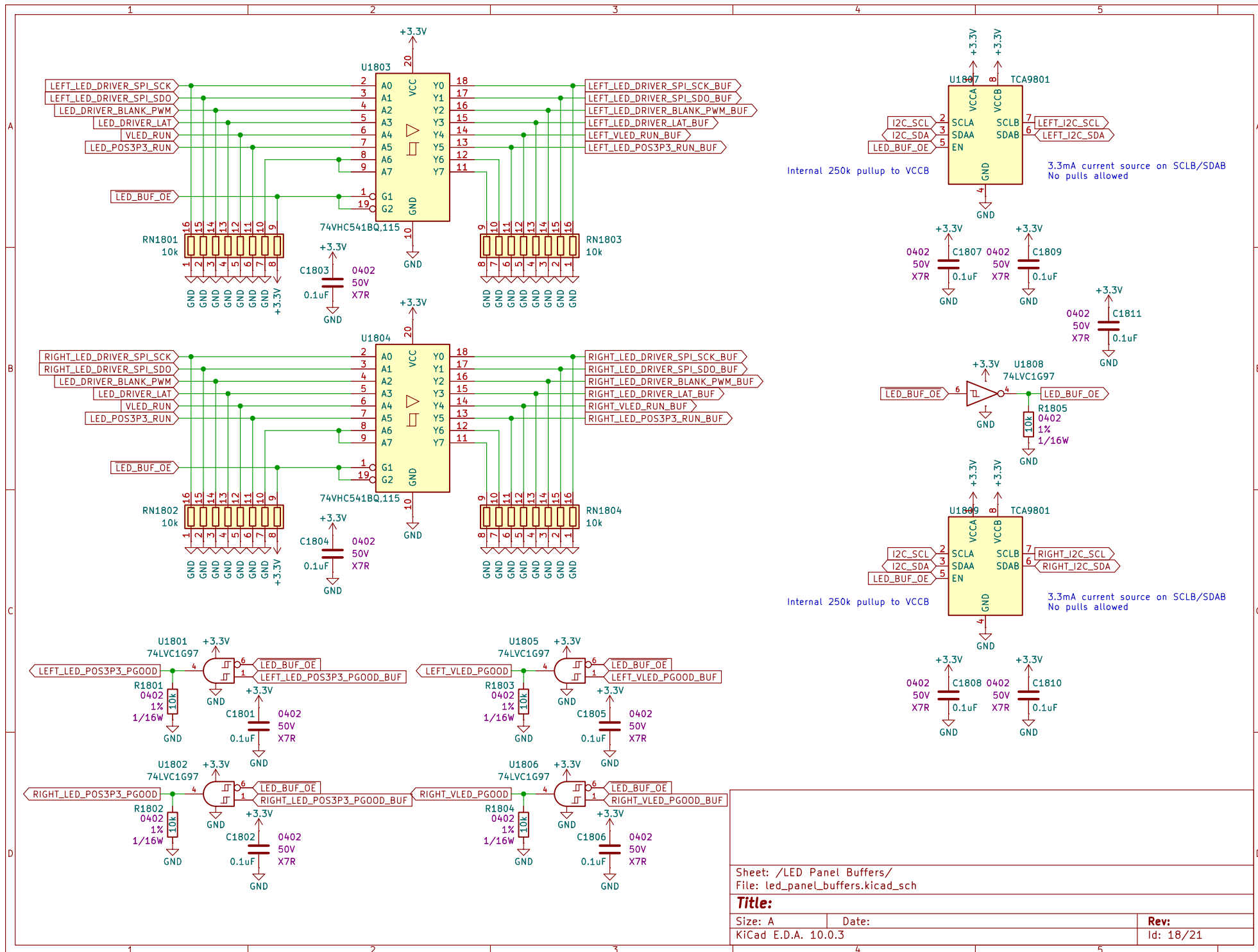


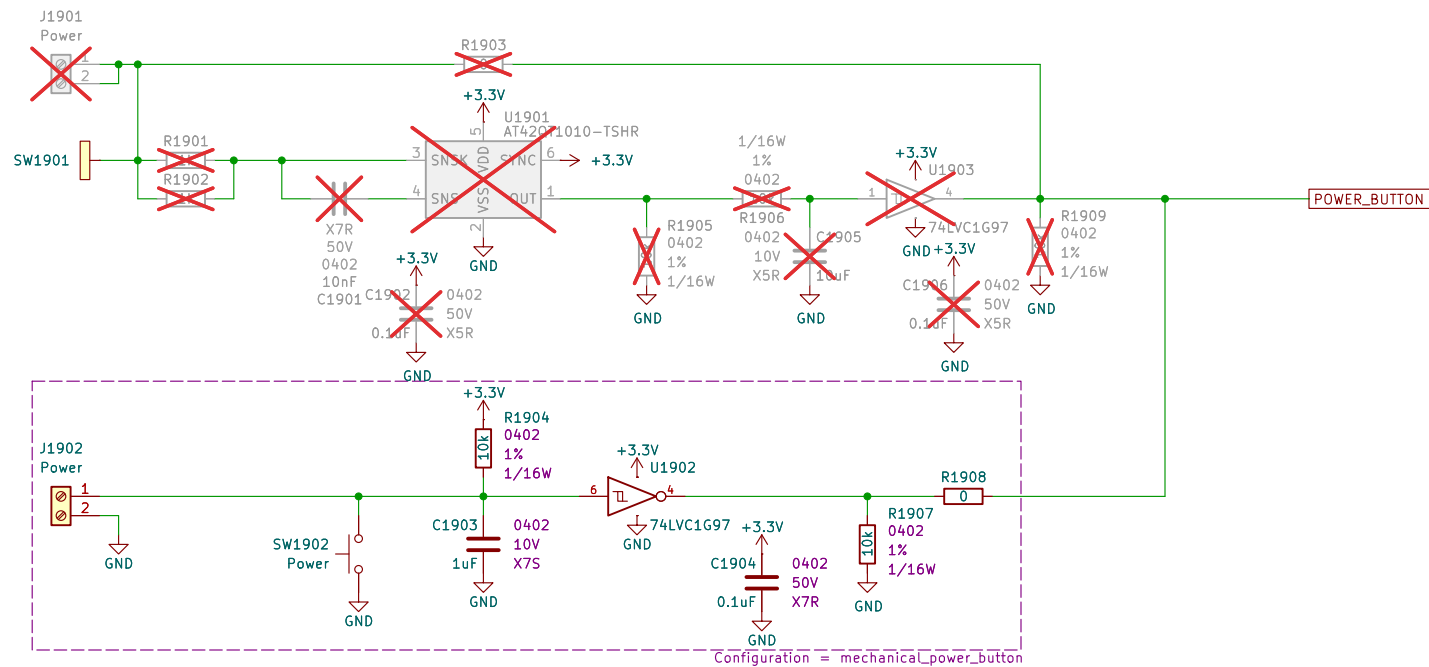
Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 17/21





Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

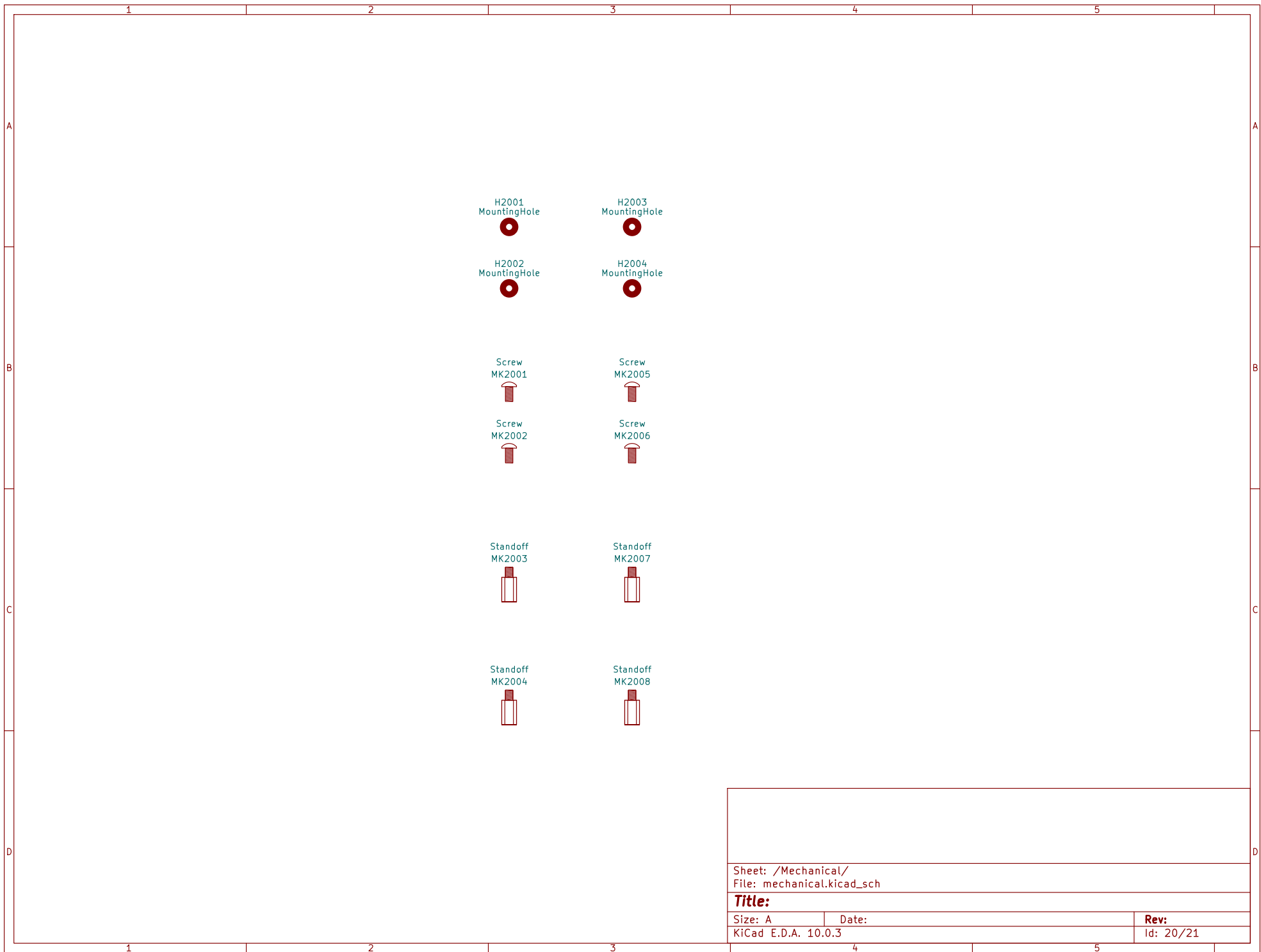
Title:

Size: A
KiCad E.D.A. 10.0.3

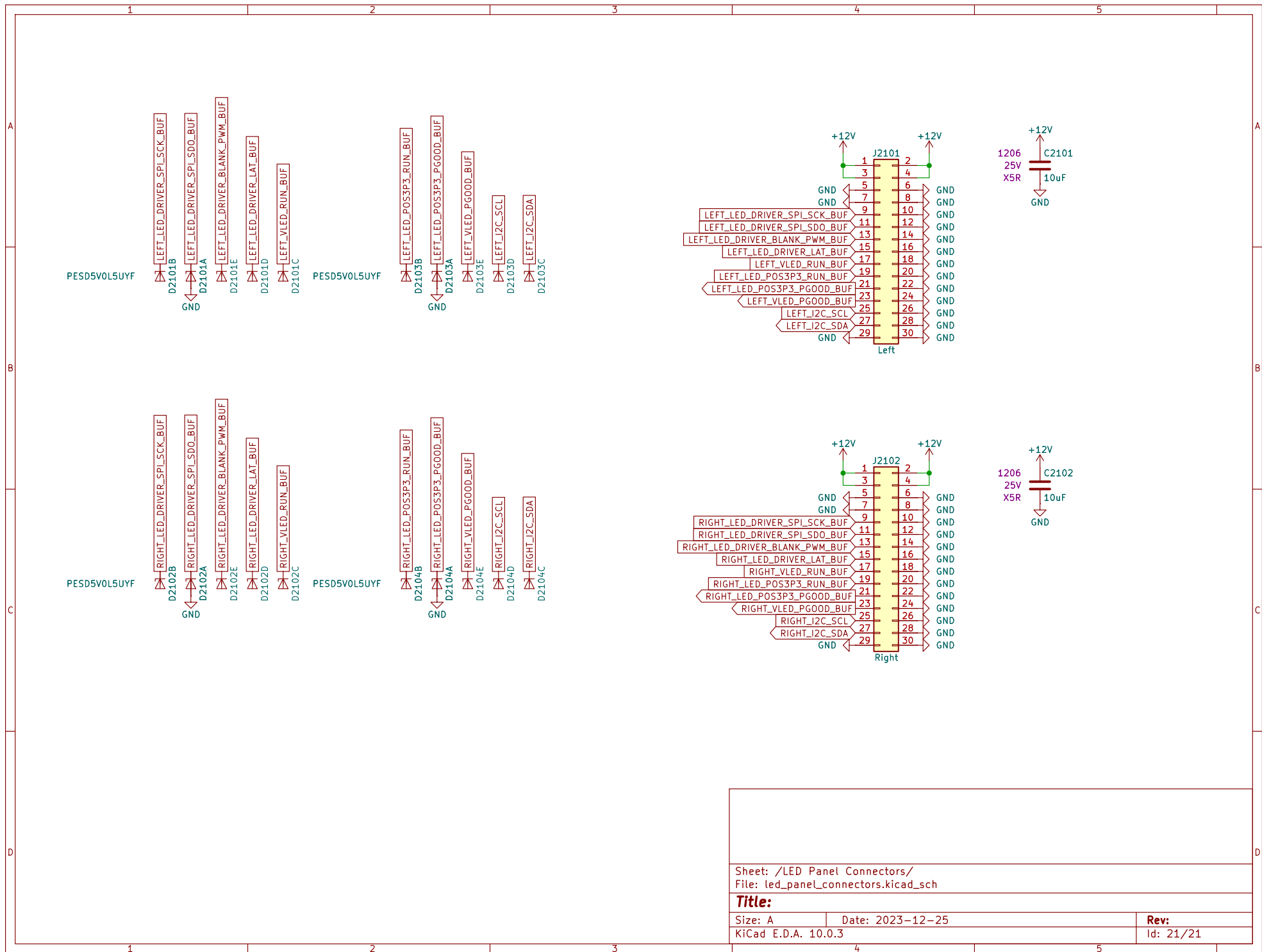
Date:

Rev:

Id: 19/21



Sheet: /Mechanical/ File: mechanical.kicad_sch		
Title:		
Size: A	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 20/21



Sheet: /LED Panel Connectors/
File: led_panel_connectors.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 21/21